

Digitalization and Inclusive Growth

A Review of the Evidence

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Abstract

This paper summarizes the evidence on the growth and distributional effects of digitalization through four channels: average productivity growth, employment and wages, access to markets, and government finances. First, digitalization has increased average productivity growth by better matching demand and supply, improving the efficiency of business processes, and boosting the accumulation of intangible capital. For developing economies, the productivity gains from “smart” automation and artificial intelligence that reduce labor costs may be lower than from the previous wave of information and communications technologies, which improved the matching of sellers to buyers by reducing search and coordination costs. Second, there is little evidence that use of information and communications

technologies has reduced aggregate employment or resulted in job polarization in developing economies, unlike the experience of advanced economies. However, distributional challenges within countries might increase to the extent that “smart” robots and artificial intelligence need complementary skills. Third, digitalization has enhanced market access for rural households, small firms, and unbanked populations in developing economies through improving information flows. Fourth, digitalization has improved the efficiency of government spending on, and revenue mobilization for, public services and welfare programs through its effect on transparency, accountability, simplification of bureaucratic processes, and adoption of new delivery models.

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Digitalization and Inclusive Growth: A Review of the Evidence

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1. Introduction

Technological progress has historically played a key role in raising global living standards. After increasing by only about half between the years 1000 and 1820, world per capita income rose by more than eightfold between 1820 and about 2000 (Maddison 2001). The second period coincided with successive waves of technological progress—from mechanization and electric power to information technologies. Without such progress, the world would be much poorer today. However, not everyone benefited equally. Differences in the adoption of technologies across countries have contributed to income divergence where, barring a few exceptions, per capita incomes in poor countries have been unable to fully catch up with those in rich countries (Comin and Mestieri 2018; Comin and Hobijn 2004). Within countries, some individuals have benefited more than others. In particular, the possibility of skill-biased technological change has raised concerns. The diffusion of a new technology increases demand for high-skilled workers and raises their wages disproportionately more than the wages of low-skilled workers, in turn exacerbating income inequality (Card and DiNardo 2002).

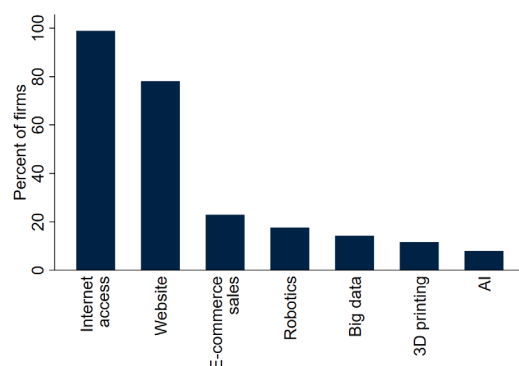
The most important general-purpose technological advancement in the past three decades has been the diffusion of information technologies. The spread of computers, internet, and mobile phones established the digital economy. There is continued progress in many areas under this first “wave” of digitalization, from advances in computer power and broadband connectivity to complementary innovations such as cloud infrastructure and online platforms. At the same time, the development of artificial intelligence (AI), advanced robotics, and machine-to-machine communication heralds the start of a second, more advanced wave of digitalization. These technologies are blurring the line between the physical and virtual worlds.¹

The world is straddling these two waves of digitalization. In advanced economies, firms widely use the technologies of the first wave of digitalization, such as the internet and websites, in their production processes (figure 1).² Yet less than 15 percent have adopted big data analysis, and less than 10 percent are using AI (figure 1). Even in the manufacturing sector, adoption of robotics averages 18 percent, and 3D printing only 12 percent. In developing economies, the first wave of digitalization has gathered considerable momentum in the past two decades. Nearly two-thirds of the population in developing economies was using the internet as of 2021, starting from practically none in 2000. Much of this increase is attributable to mobile phone connections, which expanded nine-fold during that period, and stand at approximately one per person in developing economies (figure 2).

¹ Categorizing the literature on industrial robots into the first and second waves of digitalization used in this paper is somewhat complicated. The first wave of robots was designed to simply perform tasks previously performed by workers. More recently developed, “smart” robots have the ability to make decisions and carry out tasks autonomously.

² In this paper, advanced economies are high-income countries (HICs) and developing economies are middle-income and low-income countries (MICs and LICs).

Figure 1: Digital technology adoption by firms in EU-27 countries

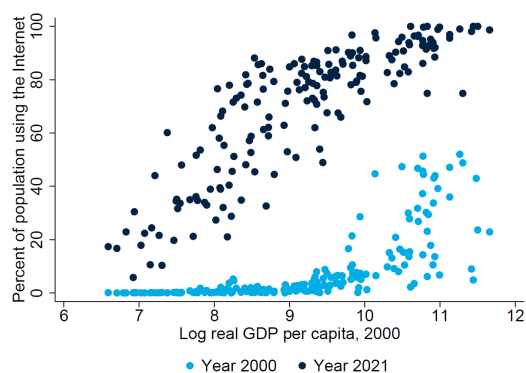


Source: Eurostat.

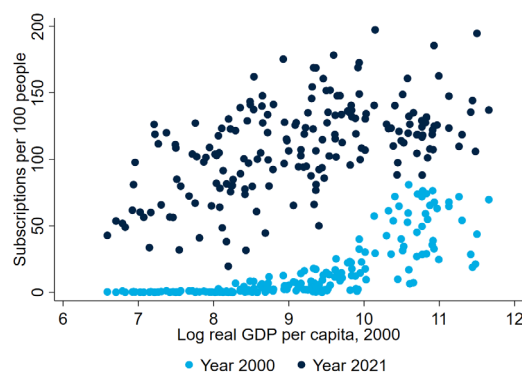
Note: Bars for robotics and 3D printing show the percent of manufacturing firms using these technologies. Other bars show the percent of all firms using the indicated technologies. Data for all technologies is for 2023 except robotics (2022), big data (2020), and 3D printing (2020).

Figure 2: Digital adoption by individuals

A. Internet usage



B. Mobile phone subscriptions



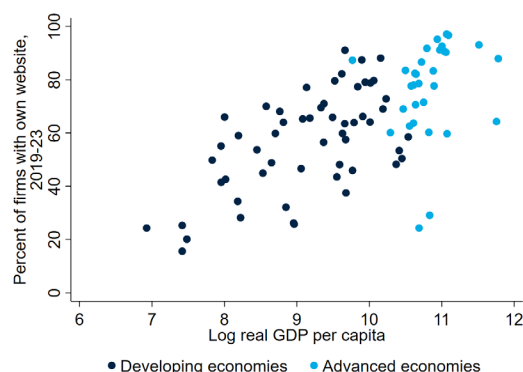
Source: World Development Indicators.

Note: In panel A, the sample includes 181 economies for 2000 and 178 economies for 2021. In panel B, values above the 99th percentile of all countries in 2021 are excluded. The sample includes 186 economies for 2000 and 193 economies for 2021.

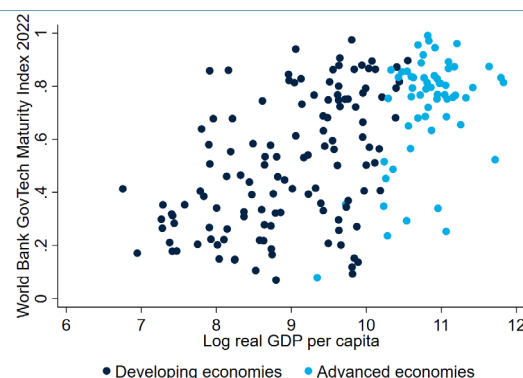
However, the first wave of digitalization is far from complete in developing economies. Although the uptake of computers, mobile phones and the internet in developing economies has progressed much faster than the uptake of previous general-purpose technologies (GPTs), the gap in the use of digital technologies between advanced and developing economies has widened compared to previous GPTs (Comin and Mastieri 2018). For example, less than 50 percent of firms in developing economies use a website, up from 26 percent in the late 2000s. In contrast, more than 82 percent of firms in advanced economies use a website, up from 55 percent in the late 2000s (figure 3a). Similarly, more than 80 percent of advanced economies have an online government e-services portal, compared to only 38 percent of developing economies (figure 3b). These and similar disparities are indicative of a “digital divide” across—and, in many cases, within—countries.

Figure 3. Digital adoption by firms and governments

A. Firms' website use



B. GovTech maturity



Sources: World Bank Enterprise Surveys; World Bank GovTech Maturity Index.

Note: In panel A, the sample includes 55 developing economies (LICs and MICs) and 31 advanced economies (HICs). If a country was surveyed more than once between 2019 and 2023, the latest survey result is included. In panel B, the index is presented on a scale of 0 (lowest) to 1 (highest) GovTech maturity. Data is for 2022.

This digital divide may perpetuate income divergence between and within countries owing to the economic dividends associated with digitalization; digital technologies boost market efficiency primarily by reducing information asymmetries which, in turn, reduce costs of search, replication, transportation, tracking, and verification (Goldfarb and Tucker 2019). However, much like previous waves of technological change, or any other economic shock, digitalization is disruptive and is likely to create winners and losers.

As the second wave of digitalization gains traction, the economic impacts of new technologies are just beginning to be studied. On the one hand, “smart” machines can boost productivity by reducing human error; optimizing complex design, production, and distribution processes; and facilitating decision-making—among other benefits. On the other hand, there are deep concerns about the distributional impacts of advanced digitalization between and within countries. Recent advances in AI and “smart” robotics could automate a large number of jobs and depress wages (Acemoglu et al. 2022). There are also concerns about competition—namely, that large technology companies headquartered in a few advanced economies and serving the entire global market, which already own large amounts of data and other inputs necessary for generative AI, have an unfair advantage over firms in developing economies in the process of creating and deploying AI.

Determining whether the poor have been more adversely or more favorably affected by technology adoption relative to the rich is an important research agenda. Yet deriving a robust empirical relationship between digitalization and poverty or inequality is far from straightforward, as surges of digital technology adoption have typically coincided with other economic shocks that also matter for poverty and shared prosperity. It is therefore instructive to identify evidence related to the channels through which the use of digital technologies can affect inclusive growth.

The paper reviews the empirical literature on the economic impacts of digitalization—in particular, on inclusive growth shared prosperity, with an emphasis on the channels of impact. Specifically, we assess the relationship between digitalization and inclusive growth by reviewing the evidence

on how the adoption of digital technologies has affected: (a) productivity growth, (b) wages and employment, (c) access to markets, and (d) government finances.³ Aggregate productivity growth is a necessary condition for poorer households to economically benefit from digitalization, as it boosts overall economic growth. However, the actual distribution of economic gains enabled by digitalization will depend on how the use of digital technologies has impacted the demand for labor, access to markets, and provision of public goods and services. Identifying the effects of digitalization on inclusive growth even through these broad transmission channels is far from straightforward. We therefore summarize the evidence around the effects of digitalization on poverty through several sub-channels within each broad category.

First, *productivity growth* is widely accepted by economists as a necessary, albeit insufficient, condition for reducing poverty and boosting shared prosperity. Differences in productivity account for half of the differences in GDP per capita across countries. In the Solow-Swan (neoclassical) model, the contribution of technological change to economic growth is reflected in growth in total factor productivity (TFP). Technological change can also raise labor productivity by raising capital accumulation. The productivity paradox of information technologies, in which the productivity gains from adoption are not immediately observable, has been a topic of much discussion. We review the empirical evidence examining three broad questions:

- Does digitalization enable better matching of supply and demand?
- Does digitalization increase the efficiency of business processes?
- Does digitalization increase capital accumulation?

Second, poor households in most countries in the world rely on labor markets for the bulk of their income. As a result, the effects of digital technology adoption on *wages and employment* matter greatly for poverty reduction. Technological change in the past has been associated with a reduction of jobs in certain tasks and industries but a more than commensurate increase in jobs elsewhere in the economy. Therefore, it is pertinent to ask not only whether digitalization has reduced jobs in the aggregate, but also whether it has impacted the demand for labor in particular tasks and occupations. The displacement of workers or a decline in wages in unskilled jobs is most relevant for poverty reduction. We review the empirical evidence examining two broad questions:

- Does digitalization reduce the demand for labor?
- Does digitalization result in job polarization?

Third, poor households are often excluded by markets. By addressing a range of information frictions, digital technologies can, in principle, improve *access to markets*. Digital goods and services—such as mobile phones, search engines, and mobile phone apps—can bring benefits to people as producers and/or consumers of goods and services by bringing prices closer to being market-determined. Digital technologies can also spur market entry and benefit small firms. The Internet and digital platforms can also improve access to financial services that is notoriously hard for poor households. We review the empirical evidence examining three broad questions:

³ This follows the framework presented in Winters, McCulloch, and McKay (2004), which reviews the empirical evidence between trade liberalization and poverty.

- Does digitalization improve the pass-through of market prices for producers and consumers in remote areas?
- Does digitalization improve market access for small firms?
- Does digitalization expand financial inclusion?

Fourth, government transfers for social security and other subsidies are typically targeted at poor households. Similarly, the efficient provision of public goods and services matters disproportionately more for poor households that can neither afford private alternatives nor “speed money.” Digitalization can affect the availability and efficient use of *public finances*. For example, it is well known that digital identification systems can enable better provision of public services and government transfers through facilitating monitoring and evaluation. Similarly, data-driven processes can improve efficiency in tax collection and customs administration. We review the empirical evidence examining two broad questions:

- Does digitalization improve the efficiency of government spending?
- Does digitalization enhance government revenue collection?

Most of the previous literature reviews on the economic impacts of digitalization are limited to advanced economies, in part due to the lag in adoption of digital technologies in developing economies. They also typically focus on a specific economic indicator—such as productivity, jobs, financial inclusion, or government revenues—and often in many cases consider the effects of internet connectivity rather than the range of digital technologies that internet connectivity enables. For example, several papers review the relationship between information and communications technologies (ICT) and productivity in large, developed economies (Draca, Sadun, and Van Reenen 2007; Holt and Jamison 2009; Brynjolfsson and Saunders 2010; Kretschmer 2012; van Ark 2016). More recently, Lu and Zhou (2021) provide an overview of the economic impacts of AI, with a focus on advanced economies. Yin and Choi (2023) review the evidence on digitalization and income inequality in G20 countries. Stanley, Doucouliagos, and Steel (2018) conduct a meta-analysis of the impacts of ICT on economic growth in advanced and developing economies. Hjort and Tian (2021) take stock of the rapidly growing research on the welfare effects of digitalization in developing economies, but with a scope limited to the economic impacts of internet connectivity.

This paper takes a more expansive approach to digitalization than previous papers, reviewing the empirical literature on the growth and distributive impacts of technologies within the first wave of digitalization (e.g., computers, mobile phones, internet connectivity) and the available literature on the economic impacts of the second wave of digitalization. Where possible, studies of advanced economies and developing economies are presented separately, acknowledging that the economic impacts of ICT may differ according to stage of development (Watanabe, Naveed, and Zhao 2015). That said, the literature on the economic impacts of the second wave of digitalization in developing economies is still limited.

The remainder of the paper is organized as follows. Section 2 reviews the evidence on digitalization and productivity growth, with a focus on the channels on whether the use of digital technologies has enabled better matching of demand and supply, increased the efficiency of business process, and increased capital accumulation. Section 3 reviews the evidence on

digitalization, wages, and employment, with a focus on the automation of tasks and job polarization. Section 4 reviews the evidence on digitalization and access to markets, with a focus on prices and financial inclusion. Section 5 reviews the evidence on digitalization and public finances, with a focus on government revenues and spending. Section 6 concludes.

2. Digitalization and productivity growth

In the neoclassical growth model, the role of technological change is reflected in the growth of total factor productivity (TFP).⁴ Technology use can also raise labor productivity to the extent it increases the accumulation of capital. Early endeavors to measure the economy-wide effect of ICT adoption on productivity used data for the United States and found little or no effect (Berndt and Morrison 1995; Jorgenson and Stiroh 1995; Loveman 1993; Oliner and Sichel 1994). These results underscored the “productivity paradox” that Robert Solow first described in the late 1980s. As the data revealed a clear improvement in U.S. productivity in the late 1990s, the narrative shifted. This is evident from the growth accounting literature and from econometric evidence at the industry and firm levels (Brynjolfsson and Saunders 2010; Draca, Sadun, and Van Reenen 2009). However, these studies neither examine the channels through which ICT use improved productivity nor consider the second wave of digitalization.

2.1 Does digitalization enable better matching of supply and demand?

The spread of the internet and digital platforms has improved productivity by enabling more efficient matches between buyers and sellers, as well as between firms and workers. Productivity gains are reflected in lower prices in advanced economies for a variety of products sold online—including books, insurance, air travel, and automobiles (Brynjolfsson and Smith 2000; Brown and Goolsbee 2002; Orlov 2011; and Scott Morton, Zettelmeyer, and Silva-Risso 2001). There is evidence of productivity gains in matching service providers and job seekers too. Based on firm-level data from ten OECD countries across four industries—hotels, restaurants, taxis, and retail trade—Bailin Rivaes et al. (2019) find that the average service provider in countries with high online platform development experienced larger productivity increases vis-à-vis the average service provider in countries with low online platform development. Evidence from job platforms in the United States and Germany suggests that online job searchers were re-employed about 25 percent faster than non-online job searchers (Gürtzgen et al. 2021; Kuhn and Mansour 2014). In Norway, the use of online platforms is estimated to have reduced the duration of job vacancies and the share of firms with unfilled vacancies (Bhuller, Kostøl, and Vigtel 2020).⁵

For developing economies, much of the evidence of ICT lowering the costs of matching buyers and sellers has been shown in studies of international trade. Based on data for 60 countries, Freund and Weinhold (2004) find that a 10-percentage point increase in growth of web hosts for the

⁴ Numerous cross-country empirical studies have found a positive association between the adoption of ICT (computers, hard disks, internet cables, and the like) and growth in GDP or GDP per capita (e.g., Colecchia and Schreyer 2002; Niebel 2018; Stanley, Doucouliagos, and Steel 2018; Vu 2011). Others have also found a causal relationship between broadband penetration and growth (Arvin and Pradhan 2014; Czernich et al. 2011; Edquist et al. 2018).

⁵ Despite the widespread availability of products on e-commerce platforms, substantial price dispersion remains due to remaining search costs, partial product information, lack of competition, and inefficient design of online platforms (Baye, Morgan, and Scholten 2004; Choi, Dai, and Kim 2018; Dinerstein et al. 2018; Einav et al. 2015; Smith and Brynjolfsson 2001).

average country contributed to a 1-percentage-point increase in annual export growth in the late 1990s, with stronger effects for poor countries than for rich countries. This finding is echoed in a subsequent study of the effect of internet penetration in developing economies on exports to developed economies (Clarke and Wallsten 2006). Osnago and Tan (2016) find that the extent of internet adoption in the exporter's country matters more; a 10 percent increase in an exporter's rate of internet adoption led to a 1.9 percent rise in bilateral exports, while a 10 percent increase in the importer's internet adoption leads to a 0.6 percent increase in bilateral exports. The diffusion of digital technologies has also enabled international trade in services, such as business and intellectual services, that were previously non-tradeable (Freund and Weinhold 2002).

Further, cross-country evidence including developing economies suggests that e-commerce platforms have allowed firms in more remote locations to access international markets by matching physically distant buyers and sellers (Hortaçsu et al. 2009; Lendle et al. 2016). Improvements in ICT have also facilitated the fragmentation of production processes across geographical locations, thereby enabling firms to specialize in a narrower set of activities and outsource the remaining tasks. Fort (2017) shows that the adoption of communication technologies (such as the internet, electronic data interchange, e-mail, extranet, or other online systems) by U.S. firms increased their likelihood of fragmenting production to other countries. There is also evidence of a positive relationship between internet usage and exports across African countries, attributable to the former reducing market entry costs associated with exporting (Hinson and Adjasi 2009).

The emergence of digital labor market platforms has catalyzed online outsourcing of IT and IT-enabled business services, a shift that has been found to disproportionately benefit workers from developing economies (Agrawal, Lacetera, and Lyons 2016). Whether developing economies can benefit from online freelancing platforms at the beginning of the digitalization process largely depends on their human capital endowment. As of 2021, around half of the world's online freelancers were based in India, Bangladesh, and Pakistan because of their workforces' strong English language and technology skills (Baldwin 2019). This pool of online freelancers may expand geographically, however, if English language dominance diminishes with the diffusion of AI-enabled machine translation (Brynjolfsson, Hui, and Liu 2019).

There is also evidence of the efficiency gains from online labor market matching in developing economies. Experimental evidence from India shows that a combination of advertising and applicant verification increased portal-based hiring by 68 percent (Fernando et al. 2023). Similarly, evidence from Ethiopia shows that lowering job application costs, by using the internet and precluding the need to apply in person, has enabled employers to attract higher-ability applicants (Abebe, Caria, and OrtizOspina 2021).

2.2. Does digitalization increase the efficiency of business processes?

Evidence from advanced economies shows that the use of digital technologies during the first wave of digitalization reduced the cost of computing across internal business functions. Based on data from 20 European countries and 22 industries between 2010 and 2015, Gal et al. (2019) find that greater adoption of enterprise resource planning (ERP) software, customer relationship management (CRM) software, and cloud computing in an industry is associated with higher multifactor productivity growth in the average firm. In the United Kingdom, DeStefano, Kneller,

and Timmis (2023) find the adoption of cloud computing is linked to firm-level productivity gains. Using data from a survey of 179 publicly traded firms in the United States about their business practices, information systems, and the analysis of information and publicly available financial data, Brynjolfsson et al. (2011) find that, on average, a one-standard-deviation increase in data-driven decision-making is associated with a 4.6 percent productivity gain.

There are also specific case studies on the effects of ICT on productivity in the United States. Jin and McElheran (2017) show how the purchase of cloud computing services improves the productivity of young companies, particularly in IT-intensive and high-uncertainty manufacturing industries. Agrawal and Goldfarb (2008) show how shared computer networks increased academic productivity at middle-tier universities. In health care, Miller and Tucker (2014) and McCullough, Parente, and Town (2016) show that electronic medical records (EMRs) are found to improve patient outcomes, and Lee, McCullough, and Town (2013) show that EMRs increase hospital productivity.

Studies from developing economies also document the productivity effects of the first wave of digitalization. Covering firms in the manufacturing sector across 82 developing economies, Cusolito, Lederman, and Peña (2020) find evidence of a larger productivity premium from using email or a website than from improving managerial capability or from exporting.⁶ Evidence from approximately 20 developing economies shows that differences in technology sophistication, ranging from manual to advanced digital processes, account for 31 percent of cross-establishment dispersion in productivity (Cirera, Comin, and Cruz 2024). For Chinese-listed manufacturing firms, Lo et al. (2023) show that digital technology boosts productivity by reducing operating costs, improving resource allocation, and enhancing innovation capability. Greater internet connectivity led to a large and significant increase in net firm entry in South Africa, notably in sectors that use ICT extensively (e.g., finance), and in the productivity of existing manufacturing firms in Ethiopia (Hjort and Poulsen 2019). Comin et al. (2022) find that adoption of more sophisticated ICT in business processes by firms in Brazil, Senegal, and Viet Nam prior to the COVID-19 pandemic was associated with higher productivity during the pandemic.

The literature has likewise produced evidence about the productivity effects of technologies associated with the second wave of digitalization.⁷ The increasing use of industrial robots in Europe, for example, is found to be associated with productivity increases at the country level in the 1990s and 2000s (Autor and Solomons 2018; Graetz and Michaels 2018). Cathles, Nayyar, and Ruckert (2020) conclude that the partial or full implementation of Industry 4.0 technologies, including robots, 3D printing, and the Internet of Things (IoT) is positively related to firm-level labor productivity, including after accounting for firm size.

If anything, the adoption of “smart” robots in advanced economies may limit productivity gains in firms in developing economies if it reduces the latter’s exporting opportunities. Whether industrial automation in advanced economies is resulting in reshoring has been the subject of a growing body of research. Artuc, Bastos, and Rijkers (2023), for instance, show that a 10-percentage point

⁶ A productivity boost associated with digital technology adoption at the firm level has also been found in studies in Argentina, Brazil, Colombia, and Mexico (Brambilla and Tortarolo 2018; Dutz et al. 2017; Ospino 2018; Iacovone and Pereira-López 2018).

⁷ There are also theoretical papers on the impact of AI and automation on TFP growth. For instance, Jones (2021) constructs a semi-endogenous growth model to illustrate that automation and artificial intelligence can contribute to long-run growth prospects by boosting TFP growth.

increase in robot density in developed economies is associated with a 6-percentage point increase in imports from developing economies, and a 12-percentage point increase in exports to these countries. Exploiting differences across countries and industries, Hallward-Driemeier and Nayyar (2019) find that, past a threshold level, the increasing number of robots per 1,000 employees in high-income countries (HICs) is negatively associated with the growth rate of the stock of outbound foreign direct investment (FDI) from HICs to lower-middle-income countries. However, Freund, Mulabdic, and Ruta (2022) find that 3D printing is, on average, supportive of existing trade patterns.

Systematic documentation of how the use of AI affects productivity is still limited, even for advanced economies. AI usage positively contributes to firm productivity in studies using data from Germany and Denmark (Czarnitzki, Fernández, and Rammer 2022; Warzynski forthcoming). There is evidence of a positive correlation between wages and exposure to AI language modeling in the United States (Felten, Raj, and Seamans 2023). At the sector level, machine learning has been found to enhance judicial output and labor productivity (Kleinberg et al. 2018), and to improve doctors' prediction capacity (Mullainathan and Obermeyer 2022). AI-based digital technologies may allow larger segments of the labor market to improve their productivity if policies support the necessary shift in occupational demand (Ernst, Merola, and Samaan 2018).

As the use of AI is still very limited in most developing economies, empirical work on the productivity impacts of the second wave of digitalization on productivity has hardly begun. One recent study of customer service agents, working predominantly in the Philippines, finds that the use of an AI-based conversational assistant increased the number of customer issues resolved per hour by an average of 14 percent (Brynjolfsson, Li, and Raymond 2023).

2.3. Does digitalization increase capital accumulation?

The literature identifies ICT investment as a key factor behind productivity growth in the United States (Jorgenson, Ho, and Stiroh 2002; Oliner and Sichel 2000). Further work highlights the dominance of IT-oriented sectors in driving the pickup in TFP growth observed in the United States during the 1990s (Jorgenson et al. 2007; Jorgenson, Ho, and Stiroh 2008; Stiroh 2002). A series of studies following similar analytical approaches have found similar results for other advanced economies (Vu 2011). The productivity effects of ICT investments have been linked to the investments in intangible capital that comprises computer-related software and data, R&D, design and other properties of innovation, and company competencies such as marketing, firm-specific training, and organization practices (Corrado, Hulten, and Sichel 2005).⁸

In their review of the firm-level econometric evidence from the United States, Brynjolfsson and Hitt (2003) argue that the surge in the aggregate productivity statistics during the 2000s reflect the contributions of intangible capital accumulated in the past. This includes complementary “organizational” investment in business process design, job training, customer service, and the like. In another review, Draca, Sadun, and Van Reenen (2009) find that there is a much larger

⁸ There are several economy-wide and industry-level studies on advanced economies, primarily using a growth-accounting approach, that find a positive impact of intangible capital on productivity growth, such as in the United States (Corrado, Hulten, and Sichel 2009); Japan (Fukao et al. 2009); and the Republic of Korea (Chun and Nadiri 2016). Firm-level studies provide additional supporting evidence (Bontempi and Mairesse 2015; Marrocu, Paci, and Pontis 2012; and Di Ubaldo and Siedschlag 2021).

impact of ICT on productivity than what the standard neoclassical growth model would predict, and it is attributable, at least in part, to investments in complementary intangible capital.

Brynjolfsson, Rock, and Syverson (2021) find that the productivity growth resulting from ICT use was initially underestimated but subsequently overestimated when the benefits of intangible investments were realized with a lag. This “productivity J-curve” can be largely attributed to the poor measurement of intangible capital in national accounts. The authors find that adjusting for intangibles related to computer hardware and software yields a TFP level that is 15.9 percent higher than official measures in the United States by the end of 2017. Analyzing firm-level data across 13 EU countries, Bloom, Sadun, and Van Reenen (2012) find that foreign affiliates of US multinationals obtain higher productivity than non-U.S. multinationals (and domestic firms) from their IT capital owing to better management practices. The advent of AI is likely to spawn a new wave of complementary investments in intangible capital.

There is some evidence from developing economies on the link between digital technologies and productivity growth through investments in ICT and related intangible capital. Analyzing data across 156 countries between 1990 and 2019, Calderon and Cantu (2021) find that broadband-capable mobile connections have an impact on economic growth through the total factor productivity growth channel, while the number of internet users drive it through the capital accumulation channel. Commander, Harrison, and Menezes-Filho (2011) identify a strong positive link between ICT capital and productivity in manufacturing firms in Brazil and India. Diao et al. (2021) show that capital-intensive technology advancements in the manufacturing sector improve firms' productivity in Ethiopia and Tanzania accordingly, with large firms experiencing stronger productivity gains. Piatkowski and van Ark (2005) show that ICT investment contributed to economic convergence between Central and Eastern European countries with the EU-15 and the United States during the 1990s. Intangible capital is found to be positively correlated with growth in China (Hulten and Hao 2012), export growth and TFP growth in the manufacturing sector in Brazil (Dutz et al. 2012), and software sector output in India (De 2009; De and Dutta 2007). In a study of 60 economies between 1995 and 2011, Chen (2018) finds that the differences in intangible capital can account for up to 16 percentage points of cross-country income variation.

3. Digitalization, employment, and wages

Most people benefit from economic growth through the income they—or someone in their household—earn from work. This is particularly the case for poor households, whose human capital is often their largest asset. Digitalization, much like any other technology, can affect people's incomes to the extent that it automates certain tasks, consequently displacing workers from certain jobs, or by placing a premium on new skills that poor or disadvantaged households often do not possess.

3.1 Does automation reduce the demand for labor?

The effects of technological change on the demand for labor have concerned society since the dawn of automation. A task-based framework assessing three channels through which digital

technologies can affect jobs and wages is a useful starting point (Acemoglu and Restrepo (2019)).⁹ First, computerization and other digital automation replaces labor with capital in completing routine, codifiable tasks, thus eliminating jobs (the “displacement” effect). Automation can also allow for more flexible allocation of tasks, thereby raising labor productivity and the demand for labor in non-automated tasks (the “productivity” effect). Finally, new technologies can also create new tasks that complement digital technologies, potentially creating jobs (the “reinstatement” effect).¹⁰ On net, the negative displacement effect of automation may be outweighed by productivity gains that increase the demand for labor, including in complementary tasks.

Evidence suggests that the adoption of ICT associated with the first wave of digitalization has not reduced aggregate employment in advanced economies. Based on data for the 2000s from 19 advanced economies, including those of the European Union, Australia, Canada, Japan, the Republic of Korea, and the United States, Autor and Salomons (2018) find that own-industry employment losses attributable to ICT-related automation were reversed by indirect gains in customer-facing industries and induced increases in aggregate demand.¹¹ In the United States, the expansion of automated teller machines (ATMs) is a much-cited example. Despite the number of ATMs quadrupling between 1995 and 2010, the number of bank tellers increased by about 10 percent between 1980 and 2010. By reducing the cost of operating a bank branch, ATMs indirectly increased the demand for tellers. Further, as routine cash-handling tasks receded, computerization enabled tellers to become involved in new “relationship banking” tasks (Autor 2015).

The few empirical studies on developing economies broadly find that the net effect of the first wave of digitalization on jobs has been positive, albeit with differences according to type of employment. Hjort and Poulsen (2019), for example, conclude that the reinstatement effect outweighs the substitution effect of automation associated with faster internet access in Sub-Saharan African countries. Studies of the impacts of ICT adoption on jobs have also investigated location-based differences. In Tanzania and Nigeria, working-age individuals living in areas covered by mobile internet experienced an increase in wage employment (Bahia et al. 2020, 2021).

Evidence about the effects of robots on jobs in advanced economies has not coalesced around a definitive narrative. In an analysis considering both the displacement and productivity effects, Acemoglu and Restrepo (2020) show that the introduction of one industrial robot per 1,000 workers has reduced the employment-to-population ratio in the United States by 0.2 percentage point and wages by 0.4 percent. A similar study on EU countries using the same density of industrial robots estimates a reduction in the employment-to-population ratio of about 0.2 percent (Chiacchio, Petropoulos, and Pichler 2018). The adverse effect of robots on employment in the EU is echoed in the findings of Carbonero, Ernst, and Weber (2020). Yet a study of seven European countries using firm-level data concludes that industrial robots have no direct effect on employment (Jäger, Moll, and Lerch 2016). Antón et al. (2022) caution against drawing clear conclusions about how employment was impacted by robot adoption in Europe between 1995 and 2015, showing that the effect is sensitive to the time period, country sample, and model specification.

⁹ The “task-based” framework followed previous investigations of “skill-biased technological change.”

¹⁰ Acemoglu and Restrepo (2018) use data from Lin (2011) to show that about half of the employment growth over 1980-2015 took place in occupations in which job titles or tasks changed.

¹¹ However, own-industry labor share losses are not outweighed by gains elsewhere.

Evidence is also indicative of the differential impacts of robot usage on jobs in advanced and developing economies. De Vries et al. (2020), using data for 37 economies, conclude that a rise in robot adoption relates significantly to a fall in the employment share of routine manual task-intensive jobs in advanced economies but not in developing economies. This reflects the fact that higher wages for low-skilled workers provide larger incentives for automation in advanced economies. It is also possible for automation in advanced economies to displace jobs in developing economies through cross-border effects. Kugler et al. (2020), for example, find that robot use in the United States has lowered employment and earnings in Colombia among workers in sectors characterized by a high level of automation in the U.S. labor market. Similarly, Faber (2020) finds negative spillovers from robot adoption in the United States on employment in Mexico.

The literature on the effects of AI on jobs is nascent. Using data for the United States for 2010-18, Acemoglu et al. (2022) find no detectable effect on employment and wages in occupations and industries exposed to AI. Felten, Raj, and Seamans (2019)—also using U.S. data—find that, on average, employment increases and wages decline in industries where AI is used. In general, there has been an acceleration in the displacement effects of automation and a deceleration in reinstatement effects during the past 40 years in the United States compared to the 40 years before (Autor et al. 2024).

Importantly, none of these studies captures data from very recent AI developments—in particular, the rapid uptake of generative AI. It is possible that the economic effects of AI will take time to fully discern in the data, just as the effects of the first wave of digitalization took time to measure. There is consensus that the advent of AI is poised to automate a wider range of cognitive tasks relative to ICT which made it feasible for machines to replace labor in routine tasks (Brynjolfsson, Mitchell, and Rock 2018). It is therefore expected that advanced economies will experience the benefits and pitfalls of AI sooner than developing economies because of the larger share of cognitive-intensive tasks in total employment (Felten, Raj, and Seamans 2021). However, much like other technologies, the gains in productivity from AI adoption could result in higher average incomes if the resulting increase in labor demand offsets the replacement of labor in certain tasks (Cazzaniga et al. 2024).¹²

3.2 Does automation increase job polarization?

A related concern about the effect of digitalization on shared prosperity is the potential for job polarization, whereby the automation of routine tasks results in rising employment of low-skilled and high-skilled workers relative to middle-skilled workers. Multiple studies provide evidence of job polarization in advanced economies. Autor and Dorn (2013) and Autor, Katz, and Kearney (2006) conclude that computerization in the United States was associated with falling numbers of middle-skilled, middle-wage jobs and increasing numbers of high-skilled and low-skilled jobs. Goos, Manning, and Salomons (2014) reach similar conclusions for Europe, where middle-wage jobs fell as a share of total employment between the early 1990s and mid-2000s. Similar conclusions are reached in studies of individual countries, including Germany, Japan, and the United Kingdom (Dustmann, Ludsteck, and Schönberg 2009; Goos and Manning 2007; Ikenaga and Kambayashi 2016; Montresor 2019; Spitz-Oener 2006).

¹² When using different model assumptions, the potential impact of AI on output growth in the literature can range from a negative impact to a continuous unbounded positive impact (e.g., Nordhaus 2021; Trammell and Korinek 2023).

Combining data from the United States, Japan, and nine European countries, Michaels, Natraj, and Van Reenen (2014) find a shift in the demand from middle-educated workers to highly educated workers in industries with rapid ICT growth. Other studies also find that shifts in demand for certain tasks accompanying workplace computerization contributed to an increase in the relative demand for skilled workers in advanced economies (Acemoglu 1998, 2002; Autor, Katz, and Krueger 1998; Akerman et al. 2015; Atasoy et al. 2013).

The relationship between the first wave of digitalization and labor market polarization is less conclusive in developing economies than in developed economies.¹³ On one hand, Shapiro and Mandelman (2021) find that a composite measure of ICT adoption (density of servers, download speeds, 3G coverage, and share of firms with websites) is negatively correlated with self-employment rates but uncorrelated with wage-employment rates. Their results suggest a possible boost to shared prosperity following ICT adoption given that self-employed workers in developing economies tend to be at the lower end of the income distribution. Das and Hilgenstock (2022) similarly find little evidence of polarization in developing economies since the 1990s, but caution that these economies, thus far, have been far less exposed to routinization of tasks than advanced economies. On the other hand, based on a sample of more than 50 countries at different income levels, Jaumotte et al. (2013) show that an increasing share of ICT capital in the total capital stock increased the premium on higher skills, thereby exacerbating income inequality. In China, Zhou and Tyers (2019) find that skill-biased technological change has contributed to rising inequality. In Africa, Hjort and Poulsen (2019) find that faster internet connectivity increased employment in higher-skill occupations more than in low-skilled occupations.

The rise of robots and AI may lead to additional substitution of capital for labor, by allowing more tasks to be automated. In a study of EU economies, Graetz and Michaels (2018) show that the share of hours worked by low-skilled workers fell with increased use of industrial robots, while the middle- and high-skilled share rose. In contrast, Maloney and Molina (2019) find no evidence of polarization, on average, in a study of 21 developing economies.

AI is increasingly able to automate prediction- and decision-related cognitive tasks that could previously only be done by people (Agrawal, Gans, and Goldfarb 2019). The diffusion of AI may therefore increase income inequality to the extent it complements high-income workers. In the United States, Felten, Raj, and Seamans (2019) find that high-wage occupations experience employment and wage growth following the introduction of AI, while low-wage occupations experience wage losses.¹⁴ Albanesi et al. (2023) conduct a similar study of 16 European countries for 2011-19, also finding a positive relationship between AI-enabled automation and employment, particularly for high-skilled and young workers. At the same time, AI could automate certain highly skilled, decision-making tasks such as medical care, legal services, and software coding, potentially reinstating middle-skilled jobs that were previously diminished by ICT usage in advanced economies (Autor 2024). One study projects that U.S. workers' exposure to AI will reduce wage inequality, defined as the ratio of the 90th to the 10th percentile of wages, by 4 percent (Webb 2020).

¹³ Martins-Neto et al. (2024) review the literature on job polarization in developing economies.

¹⁴ That machine learning systems will increasingly be able to replace high-wage cognitive tasks is reinforced by the deceleration of employment growth in abstract task-intensive occupations after 2000 in the United States (Beaudry, Green, and Sand 2016; Mishel, Shierholz, and Schmitt 2013).

4. Digitalization and access to markets

The use of digital technologies can improve market access for poor households and small firms by facilitating the flow of information that reduces transaction costs. We assess the evidence across three important channels studied in the literature. The first channel is the extent to which digitalization has removed constraints on the pass-through of market prices producers and consumers in rural areas. The second is the extent to which digitalization has enabled small firms to enter new markets. The third is the extent to which the provision of financial services through digital channels has facilitated financial inclusion.

4.1. Does digitalization improve the pass-through of market prices for producers and consumers in rural areas?

The literature is rife with evidence from developing economies on how ICT adoption has benefited agricultural producers in rural areas by improving information flows and raising the prices they receive for their products. For instance, Labonne and Chase (2009) show that improved access to information through mobile phones allowed farmers in the Philippines to negotiate better prices and selling locations, which translated into higher incomes. Relatedly, Aker (2010) finds that use of mobile phones reduced price dispersion in Niger's grain market, allowing farmers from less connected areas to sell their products at a higher price. The effect was particularly strong for products with high transportation costs. Jensen (2007) reaches a similar conclusion, showing that mobile phones reduced price dispersion and improved the welfare of producers in the fishery market in India. Evidence from Sub-Saharan Africa shows that access to price information increases the bargaining power of rural farmers, who often sell their products to traders traveling between villages and markets. In Ghana, farmers benefited from mobile phone-enabled market information services by receiving significantly higher prices for maize and groundnuts (Courtois and Subervie 2015).

Meta-analyses suggest that providing information through mobile phones increased agricultural yields by 4 percent and the likelihood of adopting recommended inputs by 22 percent in Sub-Saharan Africa and India (Fabregas et al 2019). In general, the large reduction in communication costs resulting from the use of mobile phones improved farmers' welfare in Sub-Saharan Africa (Aker and Mbiti 2010). However, the effects may vary across types of commodities. For instance, Muto and Yamano (2009) find that increases in phone coverage were associated with a higher probability of banana sales for farmers in remote areas of Uganda but not for maize, suggesting that information may be more important for more perishable products. Other studies also find that improved information flows through mobile phones reduce the uncertainty in demand for certain agricultural goods (Aker and Mbiti 2010; Overå 2006).

On the consumer side, Couture et al. (2021) find that the benefits from e-commerce for rural households in China are mainly driven by a reduction in the cost of living for rural households, rather than by income gains to rural producers or workers.

4.2. Does digitalization improve market access for small firms?

Digital platforms disproportionately benefit small firms by reducing verification costs, therefore enabling market entry. Small firms can benefit from the brand and reputation of online platforms

in the presence of asymmetric information about their quality and reliability. Rating systems that signal product quality on these platforms further enhance buyers' trust in unfamiliar suppliers. For example, evidence from advanced economies shows that better-rated sellers on eBay have higher prices and higher revenues (Livingston, 2005; Lucking-Reiley et al. 2007; Melnik and Alm 2002), and that sellers on eBay are more likely to exit the platform as their ratings fall (Cabral and Hortacsu 2010). Similarly, cloud computing disproportionately benefits small firms by eliminating upfront capital expenditures associated with hardware needs for file storage, data backup, and software programs. Using data from the United Kingdom, DeStefano et al. (2023) find that cloud computing was associated with a 13 percent annual increase in employment in young firms between 2008 and 2015.

There is direct evidence of digitalization spurring market entry in developing economies. In South Africa, around a quarter of the increase in jobs associated with the staggered arrival of submarine cables and the subsequent rollout of terrestrial fiber networks is explained by net firm entry (Hjort and Poulsen 2019). This arrival of faster internet across Africa is also associated with internet-induced entrepreneurship. In areas with internet availability, the probability of a household establishing a nonfarm business is 17 percentage points higher compared to areas without availability (Houngbonon, Mensah, and Traore 2022). Further, experimental evidence from Africa showing that smaller firms taking part in a training program about how to sell to bigger corporations, governments, and other large buyers win about three times as many contracts as nonparticipants, but only if they have internet access (Hjort, Iyer, and de Rochambeau 2020). In rural China, e-commerce platform adopters obtain significantly higher income than non-adopters, as their sales income increases significantly (Liu et al. 2021; Li and Qin 2022).

However, there are concerns around large firms benefiting more as many digital technologies require significant capital investment. Based on data from Belgium, Dhyne et al. (2018) find that large firms experience higher returns to their investment in IT-related capital investments than small firms. Similarly, using data across European countries, Gal et al. (2019) find that small firms benefit most from adopting cloud computing, while large firms benefit most from the adoption of enterprise resource planning (ERP) software. There are also concerns about competition with digital platforms, where a few dominant players can emerge with very low costs of expanding production. This means that their services can be consumed by one person without reducing the amount or quality available to others. These competition concerns are only likely to be heightened during the second wave of digitalization, as AI-related analytics are likely to disproportionately benefit large firms with access to massive amounts of accumulated user data (Acemoglu 2023; Goldfarb and Trefler 2018).

4.3. Does digitalization expand financial inclusion?

The digitalization of finance, or fintech, provides opportunities for households and businesses that are excluded by traditional financial intermediaries to access financial services (Ahmad et al. 2020; Bollaert et al. 2021; Jagtiani and Lemieux 2018; Maskara 2021). Mobile money services, in particular, have proven to have significant benefits for the unbanked poor in developing economies by addressing the cost and institutional constraints related to expanding the reach of conventional

banks (Aron 2018; Suri 2017).¹⁵ In addition, evidence suggests that alternate credit scoring based on individuals' digital presence can expand credit access to the unbanked who lack credit scores without adversely impacting default outcomes (Agarwal et al. 2019).

Several studies suggest that the use of digital technologies in financial services disproportionately benefits poor households. Asongu and Nwachukwu (2018) conclude that using mobile phones to send or receive money or to pay bills is associated with falling inequality in upper-middle-income countries. Asongu and Odhiambo (2018) also find that sending or receiving money by mobile phone is associated with falling inequality in a sample of more than 90 developing economies (except in countries where poverty is the lowest). Chinoda and Mashamba (2021) use a structural equation modeling approach to address the endogeneity concerns in the two previous studies. Using a sample of 25 African countries, they find that fintech has an indirect relationship, through financial inclusion, with income inequality. Demir et al. (2022) reach a similar conclusion in an analysis of 140 (mostly developing) countries, with the largest effects observed in the countries with the highest inequality. The importance of financial inclusion in the link between fintech and inequality is echoed by Hodula (2023).

A related strand of literature examines the ability of fintech to smooth consumption, which may help prevent households from slipping into poverty when experiencing adverse events. Evidence from Kenya, for instance, shows that receipt of digital loans and use of mobile money improves households' resilience to negative shocks (Suri, Bharadwaj, and Jack 2021; Jack and Suri 2014). The frequency and amount of migrant remittances is also relevant here. For example, mobile money users in Kenya are more likely to receive and send remittances (Jack, Ray, and Suri 2013). In Uganda, Munyegera and Matsumoto (2016) show that mobile money subscribers are 20 percentage points more likely to receive remittances from family members compared to nonsubscribers, and that the annual value of remittances received by subscribers is one-third higher. However, evidence from China suggests that although digital financial inclusion can mitigate a large portion of transient income shocks, it is not effective in mitigating permanent income shocks (Lai et al. 2020).

There is also evidence that the use of mobile money is directly linked with better labor market outcomes for poor households. Lederman and Zouaidi (2020) show that digital payments are associated with lower unemployment rates for a large sample of developing economies, building on previous work which shows that digital payments improve the efficiency of labor allocation in Mexico and Kenya (Bachas et al. 2018; Suri and Jack 2016).

5. Digitalization and public finance

Digitalization can boost inclusive growth by enhancing the provision of public goods and services and increasing access to social protection programs, which disproportionately benefit poorer households. These benefits can be attributable to the use of digital technologies that enables new delivery models, enhances the transparency and accountability of service providers, and supports

¹⁵ Studies on the economic effects of digital financial services build on a large literature on the economic effects of financial development (e.g., Beck, Demirgüç-Kunt, and Levine 2007; de Haan and Sturm 2017; Honohan 2008; Levine 2005). Khera et al. (2021) take up the little studied causal link between digital financial inclusion and growth.

the better targeting of public resources to support the poor. The use of digital technologies can also enhance the collection of government revenues—that can, in turn, finance further spending on public welfare programs—by improving government capacity to identify the tax base, monitoring and enforcing tax compliance, and facilitating tax compliance.

5.1 Does digitalization improve the delivery of public services?

Digitalization has improved access to quality public services by enabling new delivery models that overcome traditional bottlenecks in service delivery. Summarizing experimental evidence across Africa, Aranda-Jan et al. (2014) conclude that mobile phone-assisted health projects have resulted in better patient access to basic services such as medical appointments, reduced delays in communication between patients and health, lower patient travel costs, and improved patient outcomes.¹⁶

Assessing the experimental evidence on the delivery of education services, Escueta et al. (2020) suggest that computer-assisted learning improves cognitive outcomes in developing economies.¹⁷ Online learning courses, when combined with face-to-face learning, are found to be particularly effective (Acemoglu, Laibson, and List 2014; Bettinger et al. 2023). Across Africa, e-learning programs based on mobile phones have improved learners' numeracy and literacy skills when they complement in-class teaching (Aker et al. 2012; Aker and Ksoll 2019). Such use of digital technologies has improved access to quality education programs in remote areas. Bianchi, Lu, and Song (2022) find that a large ed-tech intervention that connected some of China's best teachers to more than 100 million rural students through satellite internet had long-lasting positive effects on students' academic achievement and help close the rural-urban education gap. Evidence from Ghana and India also shows the positive educational benefits of remote instruction via satellite links (Johnston and Ksoll 2022; Naik et al. 2020).

The adoption of mobile money systems can significantly reduce the costs of providing and accessing social welfare programs. In Niger, for example, relative to households that receive transfers through traditional means, households that received mobile money cash transfers are found to have 9 to 16 percent higher dietary diversity, and children in these households eat an additional one-third of a meal per day, results attributable to the travel and procedural time savings associated with receiving transfers digitally (Aker et al. 2016). Digitizing government-to-person payments has also been found to facilitate financial inclusion, by expanding account ownership among the unbanked. Paying government wages and distributing government transfers through digital accounts rather than by cash could increase the number of people in developing economies with a bank account by tens of millions (Klapper and Singer 2017).

Second, digital technologies can improve the quality of public service delivery by enhancing transparency and accountability. Evidence from Pakistan and Paraguay shows that the use of mobile apps which lowered the costs of monitoring government officials significantly reduced bureaucratic shirking (Callen et al. 2020; Dal Bó et al. 2021). The use of digital technologies to monitor public service providers has also been associated with positive outcomes. A randomized

¹⁶ These evaluations also emphasize that staff training in use of digital technologies, monitoring and evaluation of the technologies, and setting adequate incentives are critical for the success of mobile phone-assisted health projects.

¹⁷ This includes evidence from Pakistan (Beg et al. 2019), India (Banerjee et al. 2007), and China (Ma et al. 2020; Mo et al. 2020).

evaluation of an adult education program in Niger shows that using mobile phone technology to increase the frequency of communication with teachers and students improved students' learning and teachers' accountability (Aker and Ksoll 2019). There is similar experimental evidence from the health sector. In urban slums across four Indian states, biometric tracking devices in tuberculosis treatment centers that record health worker attendance and patient adherence to treatment improved patient outcomes and the quality of health records (Bossuroy et al. 2024).

Third, digital technologies can enable better targeting of public resources to support the poor by improving spending efficiency and preventing leakages. For instance, the implementation of e-procurement procedures—that reduced the costs of acquiring tender information and personal interaction between bidders and procurement officials—in the provision of infrastructure improved average road quality in India and reduced delays in the completion of public works projects in Indonesia, primarily by facilitating the entry of high-quality contractors (Lewis-Faupel et al. 2016). Similarly, digital financial platforms to monitor the transfer of funds between government agencies have been found to reduce leakage in public programs in India (Banerjee et al. 2020). In terms of the delivery of anti-poverty programs, evidence from the Indian state of Andhra Pradesh shows that the use of biometrically authenticated payments infrastructure (“Smartcards”) improved the speed and predictability of payments to beneficiaries of employment (NREGS) and pension (SSP) programs, without affecting access (Muralidharan et al. 2016).

5.2 Does digitalization enhance government revenue collection?

Digital technologies can enhance the government's ability to identify the tax base, monitor and enforce tax compliance, and facilitate tax compliance (Okunogbe and Santoro 2023). These improvements can enhance government revenue collection and enable further public spending on public goods and services, as well as social protection programs.

First, digital technologies can lower the costs to identify taxpayers by uncovering new information. In Ethiopia, for example, the use of electronic fiscal devices (EFDs) that provide real-time information transactions significantly increased value-added tax (VAT) and income tax payments (Mascagni, Mengistu, and Woldeyes 2021).¹⁸ In West Bengal, India, the demonetization policy in 2016 resulted in a higher reliance on electronic payments, in turn increasing the amount of sales firms reported for tax purposes (Das et al. 2022).¹⁹ In Ghana, tax collectors using electronic tablets collected 103 percent more tax revenues than collectors not using the tablets (Dzansi et al. 2022). More sophisticated digital technologies, such as machine learning, have helped the authorities in Sierra Leone to determine the taxable value of properties (Grieco et al. 2019).

Second, digital technology may be used to better monitor and enforce tax compliance through the automation of validating self-reported tax liability with other data sources. In Rwanda, 43 percent of taxpayers over-report or under-report their VAT sales compared to what is automatically recorded in electronic billing machines (Mascagni, Mukama, and Santoro 2019). In China, the introduction of computerized invoices is found to have had a large impact on public finances,

¹⁸ However, Mascagni, Mengistu, and Woldeyes (2021) also show that the increase in VAT sales as a result of EFD adoption is offset by an increase in deductible costs that substantially reduces the potential revenue gains.

¹⁹ Evidence from Uruguay shows that VAT rebates for electronic payments increased non-cash transactions but did not provide significant additional information on firms outside the radar of tax authorities (Brockmeyer and Sáenz Somarriba 2022).

explaining a cumulative 27 percent of VAT revenues and 13 percent of government revenues between 2002 and 2007, achieved primarily by making it more difficult to falsify deductible claims (Fan et al. 2018). Digital technologies can also support audits that identify non-compliance by enabling the analysis of relevant taxpayer characteristics, such as business activity, quality of record keeping, past compliance behavior, and discrepancies from third-party reports (Khwaja, Awasthi, and Loeprick 2011; Loeprick and Engelschalk 2011).

Third, digitalization can facilitate compliance by simplifying taxpaying processes through e-filing systems and e-invoicing systems. In Tajikistan, the introduction of e-filing reduced the time spent on fulfilling tax obligations by an estimated 40 percent (Okunogbe and Pouliquen 2022). In Uganda, a new e-filing system doubled the number of presumptive taxpayers and increased revenues, especially when coupled with a one-stop shop that furthered lower taxpayers' compliance costs (Jouste, Nalukwago, and Waiswa 2021). However, the success of e-filing systems in reducing tax compliance costs depends on the policy environment, including the ability to generate sufficient public trust in e-filing systems (Yilmaz and Coolidge 2013). Digital technologies can also improve the communication between the tax authority and taxpayers. For example, digitally transmitted messages from the Rwanda Revenue Authority aimed to encourage compliance increased tax payments far more than physical letters (Mascagni and Nell 2022).

The rise of the digital economy has also brought challenges to the traditional tax system. This includes difficulties in collecting corporate income taxes from large, global high-tech firms (Ting and Gray 2019), allocating taxing rights on income generated from cross-border activities across countries (OECD 2015; OECD/G20 2021), and collecting value-added tax and goods and services tax on the growing volume of online purchases by private consumers from foreign suppliers (OECD 2018). Digital technologies might themselves provide some solutions, such as in the case of reducing cross-border tax fraud (Kitsios, Jalles, and Veridier 2022).

6. Conclusion

This paper summarized the empirical evidence on the economic impacts of two waves of digitalization with a focus on inclusive growth. In doing so, it focused on four channels of impact: average productivity growth, employment and wages, access to markets, and government finances. We can draw four main conclusions for developing economies that distinguish between the growth and distributional effects of digitalization, highlight differences between the first and second waves of digitalization, and consider differences in the implications of digitalization for firms, households, and governments.

First, the average productivity benefits of digitalization can accrue differentially to advanced and developing economies based on the channel of impact—the matching of demand and supply, the efficiency of business processes, and capital accumulation—as well as on differences between the two waves of digitalization. The diffusion of ICT during the first wave of digitalization matched sellers in developing economies to buyers in advanced economies by reducing search costs. The use of ICT in business processes also enabled the remote coordination of complex tasks at a low cost, which resulted in a new wave of labor-intensive export-led growth in developing economies as multinational firms combined high-tech ideas with low-wage workers in developing economies through global value chains. However, “smart” industrial automation and AI could make such

cross-border production sharing less profitable by reducing the relative importance of wages in determining competitiveness. Furthermore, the productivity benefits of digitalization through the capital accumulation channel derive mainly from intangible capital. Such assets put a premium on innovation capacity, skills, and organizational practices which typically lag in developing economies.

Second, there is little evidence so far that diffusion of ICT has either reduced aggregate employment or resulted in job polarization in developing economies unlike the experience of advanced economies. This is explained, at least in part, by the lower level of ICT adoption in developing economies. But there is also the question of skill composition. The automation of middle-skilled jobs in advanced economies was less relevant in developing economies that were dominated by low-skilled jobs. As the second wave of digitalization gains momentum, there will be new challenges. Investing in “smart” robots and other forms of industrial automation is likely to be orthogonal to the relative factor prices in developing economies that are relatively abundant in low-skilled labor. AI can dampen inclusive growth to the extent that it is associated with skill-biased technological change, but it may also improve inclusive growth by substituting for advanced skills that might be lacking in developing economies.

Third, there have been several gains for poorer households in terms of access to markets. Evidence from developing economies shows that people living in remote, rural areas have benefited through higher prices for agricultural products and lower prices for products they consume. Unlike technologies traditionally embedded in heavy machinery and equipment, digital platforms and apps—facilitated by personal computers and mobile phones—are not scale-intensive, thereby creating opportunities for smaller businesses to enter new markets. Mobile money platforms have enabled previously unbanked populations to gain access to low-cost financial services. Differences between the first and second wave of digitalization are minimal here with the exception of access to markets for smaller firms that may be disadvantaged by the use of automation and AI which are typically scale-intensive.

Fourth, poorer households can benefit from the use of digital technologies in supporting the provision of public services and social welfare programs. Evidence from developing economies shows that digitalization has improved the efficiency of public spending and revenue mobilization primarily through its impact on monitoring, transparency, and accountability; the simplification of the bureaucratic processes; and the adoption of new delivery models. Differences between the first and second waves of digitalization are minimal here, although the widespread use of AI-powered technological change is likely to be contingent on ICT use. For example, the use of big data analytics and machine learning by governments to improve the efficiency of tax collection and public spending that enable redistribution will likely be negligible in the absence of basic digital systems.

Making the benefits of digitalization described in the paper more widespread is dependent upon the greater use of digital technologies that is far from universal across developing economies. As a result, the question of identifying policy levers that can increase the adoption of digital technologies is important. This paper does not review the evidence that attempts to link to policy choices to the diffusion and economic benefits of digitalization. Investments in internet connectivity are a pre-requisite, but even where a baseline level of internet connectivity is

available, usage of digital technologies is not always widespread. This may reflect the absence of necessary analog complements that range from regulations that enable firms to leverage digital technologies to compete and innovate, improved skills of firms and workers that place people in a position to take full advantage of digital opportunities, and accountable institutions, so that governments respond to citizens' needs and demands (World Bank 2016). Digital technologies can augment and strengthen these complements, as seen in the case of financial inclusion and the efficiency of public spending.

There are many directions for future research to add to the evidence base. For one, there is a clear paucity of literature on developing economies, even regarding the first wave of digitalization. On average productivity growth, there is little evidence thus far on whether the use of ICT has accelerated income convergence or divergence across countries. On distributional issues, the set of studies analyzing the impact of computers, mobile phones, the internet, digital platforms, apps, and other software on jobs in developing economies is particularly thin. This may be attributable, at least in part, to the lack of data that track businesses over time. On the impacts of the second wave of digitalization powered by AI, while the literature even on advanced economies is nascent, opportunities for leapfrogging in developing economies could be a fruitful area for conducting case studies. In general, there is also need for greater discussion on whether the use of digital technologies is always consistent with factor endowments and factor prices, especially in developing economies where skills may be in short supply.

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